Active InferAnt Stream 011.1 ~ April Agents Bring May Simulations: OpenManus, RxInfer.jl, and ...

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all right welcome everyone it is April 2nd 2025 and this is active stream number 11.1 and I'll begin with a GitHub push looking forward to everyone's updates and ideas and questions in the chat this should be an interesting Stream So the title is April agents bring May simulations open Manis RX and fur. JL and agent laboratory I have a few things prepared a few software implementation we can play around with also some topics to explore not sure how any of the demos will go live however it's the hope that a few things will be pretty interesting and we can fix and explore along the way so April agent Spring May simulations or as Alexander just pointed out April agents may bring simulations there are many exciting developments and advancements in the multi-agent synthetic intelligence systems space right now in a pretty broad sense shortterm medium-term all the terms let's explore a few articulating complimentary interesting kinds of multi-agent Frameworks autonomous Frameworks look at different ways of looking at what makes agentic what kinds of systems engineering are being done within and among the agents all these kinds of orchestration questions looking where we're at instantaneously in April 2025 and more broader so um there will be some jumping around since some of the pieces take different amounts of time to to load or play out so again I'm not sure what exactly will come through that'll be part of the fun though um on all of these branches I've been having some interesting times there are three cursor windows open so starting with the two llm type models right now is running in this terminal the work of Samuel schmidgall agent laboratory I'll read the overview agent laboratory is an end and autonomous research workflow meant to assist you as the human researcher towards implementing your research ideas agent laboratory consists of specialized agents driven by large language models to support you through the entire research workflow from conducting literature reviews and formulating plans to executing experiments and writing comprehensive reports this system is not designed to replace your creativity but to complement it enabling you to focus on ideation and critical thinking while automating repetitive and time intensive tasks like coding and dentation by accommodating varying levels of computational resources and human involvement agent laboratory aims to accelerate scientific discovery and optimize your research productivity um this has an Associated paper describing more so I got it to work initially output a really awesome 17 page paper images equation I have the PDF around then I started modifying it trying to get more logging get more access to it uh and kind of went too far and now I've just pulled back to the main branch that they've provided and just to show this experiment while this one's running and then we'll see how long it takes hopefully it'll finish within the stream um this yaml file which they provide some initial examples of I made a variance that um has the following sections copying it over with the control I know write a new Yo compliant file that has this semantics the goal is to develop a partially observable marov decision process PDP implementing active inference that is in a thermal homeostatic setting so the kind of the classic air conditioner do nothing heater temperature the API key for open AI is put here and the uh model is provided for the llm backend and the review I guess separately language here's different stages so I found that when larger numbers were used there were some interruptions I don't know if that was because of stepping away from the computer so these are relatively low levels of steps but it'd be conceivable and interesting to see what happens when literature review is huge numbers or or what happens there how to use a structured knowledge base for literature review um it's also possible to run multiple labs in parallel then there's these solvers steps I wonder if that will throw an air later we'll see um these solver steps which are described in the mle solver and the paper solver load an existing save and to compile all the way on through the latc PDF output and then this copies over the notes format and it's doing a pretty multi-staged effort uh the package setup did have a few other packages that on my computer I needed to set up but was pretty easy to get through those with helper cursor 2 and get uh the virtual environment and at least get it along the way using the API key so this is pretty cool and uh again keeping with this sort of just snapshot there I'm I'm sure with slightly different setups or just different settings these are all really functional I'm just showing how they're running for me at this moment and uh hopefully coming across some useful outputs um but big picture agent laboratory really cool and partnering it in this sort of multi-agent autonomous agent llm type coordination uh in in the second window we have open Manis so I'll read the description manace is incredible but open Manis can achieve any idea without an invite code it's a simple implementation so we welcome any suggestions contributions and feedback enjoy your own agent with open Manis so look at the uh demo here's videoing into their setup describing the task basically carrying out this task within the repo that it's running from so here um there was I I started with a request write a research report on Davis California and I output this after 20 steps but it popped up a browser um and now running a enhanced version to see if it's possible to get a longer report um so it could just be how I'm running it or setting it up it could but just sort of conceptually we'll look at the architecture of how it's set up even if nothing to impressive comes from it and uh those will be the two llm type [Music] um processes running here we are back in the PDP research review writing code writing um we'll see if we if we can tell what the output but there are subtasks um to what extent these are explicitly described or to what extent there's an initial uh plan or hierarchical plan formulation how many steps that's taken in how General the prompting all those kinds of things a lot of degrees of freedom just um seeing how it works here is just you know interesting and then also again open manace where here let's start this one again with a new but did it even result let's try to improve the logging on open Manis then we'll go to the third window which is the RX infer do jail given this type of quote successful output we ensure we are capturing all outputs of tool use better and enabling running tools more intelligently in order to achieve sequential incremental improving outputs General case all right and then the third window is going to be the RX and fur. JL example Bulls repo so here's the open Manis now in the third window will be a par repo for RX and fur which is uh under some very interesting and good open- source development as part of this growing interesting RX andur ecosystem including also recently RX environments not going to be probably covered here and also this recent RX infer server so opening up the possibility for different languages to use API calls into these RX infer type pipelines through their their server and uh this might connect with some general topics and multi-agent just depends on what people write in the chat and where these tools sort of take us okay using cursor 48 using for the most part Claude 3.7 agent mode all right started with the GitHub push and let's just check on each of the uh ongoing language model uh then look to some of these recent examples from RX and fur including how they integrate with neural networks and see if we can generalize that or connect that also to these llm type methods um and yeah see where the different work products from these more llm tool use agents where that gets us and where we get with okay here we go yep Mily Max steps let's just do one Max of everything so that it happens from all the stages okay I'm going to um make a new API key to put into this moving it off screen so I can put in this API key this way and save the file so that'd be one way I thought of modifying it having a uh environment file or other method for the API lot of flexibility with which llm and how to call it so that could just be one thing and then here's okay so let's run that again AI lob repo is the script and then it looks to the location of the Amo file so AI lab repo all right looking at it while it sets up gives a different language default model maybe overridden um the laboratory workflow is a 400 line 500 line method the archive agent here's the argument that parses uh or argument parsing for the yaml parsing AML and then main for the main logic of this main script that's now playing out here sets the llm backbone determines whether it's in human mode or not we can try that look at the Amo and flip that value compile PDF go from the latac pl text all the way through at least trying to compile the PDF uh load in previous workspace if there are parallel Labs what to do in exception uh failure whether there's a archive research agent and the indices of the lab then there's those limits on each of those steps number of times to iterate through these different steps um and how to parallel getting the API keys from the Y whether the open AI key or the Deep SE key has a logic for if it's in the human interactive type mode uh asking questions or whether it uses the uh yaml information language human and loop mode possibility for different stages to have different human in or on the loop agent models which llm to use for phases here it looks like this is a set sequence of phases it could be specified in a ton of different ways uh parallelism instructions some exception Clauses checklist practical make a better config system yaml advancements make the ability to have agent build on top of the research run agent labs in parallel not sure if that's to do or what was done um so this is going to keep running hopefully we will not hit that same error um should be a little faster but it still might be a couple minutes on that one all right now to this one so something was happening in the open manace um but we asked cursor in the agent mode to enhance our ability to to understand because the the um the research report was not very long so maybe just it's handled differently with the database and the shared State and and this is an example of kind of a classic and and uh durable type of tradeoff in multi-agent modeling different kinds of information uh interactions ranging from this just kind of informally implicit multi-agent Hub and spoke type computational architecture um if not Bas graph where there's a niche that different agents read and write from and this can be uh in certain digital settings and even probably some physical ones too describe the uh possible uh layouts of how to organize shared State uh in cont cont with where there can be direct interactions like interactions among nestmates as well as with the shared environment and in the context of uh doing these computational agents it is like what is being transi what is being remembered and modified how where's that statistical updating of a probability distribution like the RX and fur side of things um whereas it some other kind of model update where is it context for a language model or um to what extent is is that update private within that agent uh carried like sort of like a nestmate level memory to what extent is it accessible by different processes and it kind of just ends up um being just in uh some particular way that's very situational so this is just showing a few of the tools and hopefully with people's uh comments or or suggestions let's just see how many minutes it takes to look into a few different functions or at least connect H how certain uh package analyzes this connect that to um uh ways we can integrate amongst these different Frameworks so okay now we have um agent laboratory is running in the first one open Madness we ask for these improvements now run the improved open Manus that so sometimes there's a really obvious way or an entry point um and I think good code-based design can go a long way I know there's a lot of cool work being done with the cursor rules and the MDC files a as well as the mCP server calls those are all things to uh look into but even at a a code base of this level sometimes it's a little bit hard to even know uh which which script to run or which entry point so sometimes it's even useful just to ask in the agents on the right side to even write that as it as it needs to be and also to update that experience like okay I don't want to do any terminal uh extra arguments just make it so that this works um so let's let cursor work on that while we look on to third of the three um modeling approaches the RX and fur Okay so the RX infer examples repo is maintained by the reactive base organization um lazy Dynamics fellas and what I have in the doxology Fork is a another folder called support and right now it has a few two scripts there's setup. JL which is at least a starting point for setting up the packages and and gives a a scaffold for adding other pack packages that you might want to install or something like that and then importantly the notebooks to scripts script so this can be run from here with Julia notebooks to scripts the file name Julia space the file name and takes the documents that are in the original repo the example folder there's different um folders within examples but in the uh this script converts those into a matching scripts folder so we will look at uh two or maybe more of these script uh analoges of some RX and for examples um they're a little bit more flexible to work with it's possible to break them out into multiple components and some of them probably run a little smoother in the notebook setting uh but by and large they uh all to with just a a few tweaks or ordering changes so we'll look at some that that are already uh some variants have been made of and then just drone or um some other interesting one let's let's explore it okay agent laboratory continuing um cursor Improvement of open Manis proceeding on this side so it's like the their agent orchestration their micros service um process orchestration all all all these really Advanced um services and software perplexity Etc pretty interesting how they how they do it and surface it so we leave that going and wait for it to complete while looking at the RX andur examples so um first example will be uh recently written recurrent switching linear dynamical system this one uh where we're getting to with it just to sort of put the um and this this is not not to say that this is perfectly statistically optimized just to show some of the expression capacities of where where we can model and then from there figuring out um is rx infer where where might it be at different stages in different projects ranging from uh a uh statistical power type systems modeling intuition skeleton scaffold probabilistic layer all all the way on through complimenting and Inter integrating uh with different machine learning pipelines like where where and how my and what if probabilistic modeling in in fusion with some more dynamical systems time series modeling neural networks llms uh all these different architectures of statistical models it it's um where how do we compose useful systems for variously more on the recognition Direction doing a latent State inference from noisy data which is a commonality of of the models that that will be looked at so in this um you or using it to recognize latent States from noisy data Andor generating predictions from Laten State estimates so here in this setting there's a a noisy sequence of measurements and at a a discrete layer there there's a switch between two regimes one and two where the second discret state has this higher amplitude oscillation and so there's a switching between these two modalities and we explore how hard is it to have a three-state switch model um so at the sort of faster layer there's this interpolation uh Auto regressive type model which we can look at with the app model statements and then at the slower level there's this linear um and then the other example that has uh a lot to do with this first example is um uh three-dimensional in time Loren attractor with noisy sampling coming from it and in this uh example integrating neural networks with flux JL there's a two-step process that has a lot to do with the continuous time time series interpolation and the slower switching model here at the first level there's the learning of a a a statistical model of the Loren attractor State space using a flock neural network in Julia then probabilistic parameters according to a at model statement in RX and fur are used to model certain distributions of that train neural network so it's like a common theme is so here's now the three dimensions of the Lorenz tractor so the blue line is the inferred from the actual basian posteriors uh reflected by these matrices and their change through time um in terms of now possibly miscalibrated it's not to say that this makes this an accurate assessment and I think there's a lot to um to to look at there but these are posterior estimates that in certain conditions or situations or data sets can give give better estimates and other Alternatives that have for example in this situation on this dimension for the lorens is a uh parametric estimate in terms of a mean and a variance of this posterior distribution constrained to the gaussian family of just mean and variance parametric statistics uh rather and that can be computation Ally implemented typed simulated Etc in in a number of ways with an explicitly distributional analytical nature knowing that it's going to be this normal distribution now if the underlying state is is bodal then there are all the sorts of attractor States depending on where it goes and how the optimization unfolds like whether it over disperses to capture the mean or whether it sharpens to go onto the median like a local Optima so but all that being said it gives this parametric estimate in terms of an explicit mean and variance which can be used in any statistical method for what would otherwise be distributed throughout the weights of like the neural network with many parameters needing to be simulated for inference whereas with these more distilled summary statistics on the posteriors it can be compressed to a very small um scale even though as as uh more parameters are added it the fit might be better because interal Network might be able to to know that there was some more structure than just the the two parameters that allowed to be specifying this distribution so that's kind of the um topic that I think in in a lot of ways regardless of what the agent orchestration layer is which this is giving us some windows into there sort of this orthagonal analytical or behavioral scientific framing of this information environment using uh using probability distributions to model outputs of other models some places in motifs where that could be like parameter on parameter on parameter or it could be llm on llm on llm all these different chains of uh of interactions between models that are more like let's have a ton of parameters and try to interpolate a function and pick the size of the model out of convenience or or compute like why 7 billion parameter language model because maybe uh just in the short term it's easier to fork or use the the training methods that are known or whatever it may be but use a certain given large number of parameters choose your own or some other common number and and then hope that there's enough degrees of freedom and in the curriculum of the training that it's it's uh that it interpolates and learns the space and does the good function approximation and and you don't have to do an analytical description of like the information space yet it can learn a lot of the important structural features um it whereas for example a explicit probability distribution over Words which is kind of has a functional encoding in the the larger um many parameter type like llm type learn all the statistical associations including higher order ones among different textual inputs and then learn the the specifics of that from from the training data was but spe letting those weights and coefficients amongst all of the billions of parameters become fit through the training uh in this app model type RX infer um setting and then also in in the textbook group there was um one person was interested in posture so we modified it so maybe we can modifi another of gray is the proception noisy signaling coming from the body then there's the perceived and actual posture embodying the the uh switching and constraints chaotic uh dynamical underlying system with a lorens being like the transition between two different modes this could be like seated biomechanics and standing biomechanics um or different uh the the Loren or other attractor systems could be sampled from uh or or mixtures of systems could be uh model multiple ways of modeling the generative process that's giving rise to these noisy observations doesn't have to be just fit with um but similar theme um let's look at the app model that defines the probabilistic part of it here's the state space model which can be uh in certain places and ways run in situations where the larger neural network might not necessarily be able to operate like it all the variables that have to be stored within a semantics of this being the arguments passed to this SSM model the can be interpretable true variance posterior estimates um explicit composition of different statistical distributions um the there's more to it for uh the for logging and for coordinating the calls and the orchestration of how these app models get used in The rhythms of of other API calls and and program Logics but the app model is kind of the this statistical kernel that is the one that gets parsed and and rendered and and uh used as the underlying declarative base graph for the RX Ander model that then gets a certain scheduling of either the standard sort of Passover all of the nodes and update them in this order uh imperative style update this and then do this and then do this and then do that and then to what extent that that that scheduling of updating strategies on that base graph that is what reactive message passing in principle kind of opens the degrees of freedom around so it's like well this can be under this timing this one can be triggered by this or react to this ra rather than the whole graph being either updated sort of in a uh sweep or just in any given fixed um for Loop type computational execution uh approach to that distribution so this is the joint generative model of the state space model inferred from from the neural network so the of the uh from the noisy Loren data so um let's check the other okay agent laboratory still writing cursor finished commands to run to run and log the demos written hello Zig train the neural network on the Loren's noisy data so using just a ton of parameters throwing it at the problem training the neural network okay test sequential tools this is what we get there should be at full function logging of the vest sequentially improved actual plans documents research outputs Etc logged and enacted by this system agent laboratory still going okay that was the neural network learning the neural network and then describing parameters of a um of a uh specific app model so those were two of the um but let's look at a new one like drone Dynamics all right so hav't done this one let's see what it what it results let's let's see what it looks like so this is in advanced examples drone Dynamics notebook drone do GI all right awesome drone dog GIF 2D drone drone 3D multi looks awesome so if we can recapitulate this and start extending it a little bit and understand the app model add in some logging look at some other visualizations um adapt it to foraging and the honeybee or something like that like there's there's so many um so this is just as as we watch these three processes play out which again under under total no May make it happen and do it comprehensively now these you know two weeks later the level of function could be I guess somehow deteriorated in some way but considering just for the most part largely improved a lot of the things that I'm doing certainly someone with different or or more development experience Andor another more advanced version of cursor that could just iterate a few more times uh or a little bit better with coding so it's it's pretty wild um and and I'm I'm excited to see like what people do with uh it's not just Vibe coding it's it's more different than that um I think in in ways or at least it can be okay quick scan over the Drone example these examples demonstrate the use of RX infer for motion planning the animation show the inferred trajectories from probabilistic inference rather than simulated executions for more realistic simulations especially in the 3D drone example the model would need to be extended with reactive environment # RX environments that responds to the drone's actions during plan execution if you're interested in collaborating on a more realistic implementation please open a discussion and let's work on it together and that links to to a discussion on reactive base okay let's quickly look through it import packages environmental features like gravity physics of the mass inertia radius and force limit properties of the drone aerial properties of the Drone state the Drone Vector of these variables model specification location State transition based upon this declarative statistical style graphical model specification for the Drone model that's going to get rendered up and done all these ways with the RX infer using the app model macro planning Loop included inside of the app model Macro for the time Horizon involving something like rolling out the prior onc consquences of action physics related with the state transition s but how does it select action then uh probabilistic inference not sure what unscented means here but some kind of state transition approximator this is an interesting function I I think it'll be like is it really this easy to make a wrapper function called move to Target if so that feels like a really high level function even for a 3D nav system to to be able just to be like move the Target and have it have this um definiteness that that's that'll be interesting to look at how that happens uh initial condition specification plotting of the Drone animation of the Drone so hopefully getting the script running will get us to understand um all these different visualizations and maybe make a few new visualizations just using the script and and generalizing it from there let's check on the llms then continue with the Drone all right agent Laboratory continues to use this um we had some really fun interesting discussions in the Discord so like just taking a little snippet here these are llm costs um postto let's take a fresh perspective in developing our PDP model for thermal homeostasis focusing on agile design that prioritizes user Centric features while maintaining strong mathematical Integrity here a newly formulated plan that emphasizes practical user engagement and theoretical robustness three action states of the PDP five continuous band uh on a on a Continuum making a discrete State space 5 option Laten space so it's laying out different aspects certain parts of it are repeating things it could just be logging it could be could be other ways to do the prompt engineering like this just looks like it's repeating things right now maybe it's actually asking if we're ready or something but it's yeah I I think that'll interrupt let's see okay something is happening let's leave that highlighted return to it here looks like the cursor agent make the updates and run it and fun it okay um back to the Drone um so we looked at the readme for the Drone this is in the scripts folder which basically takes this The Notebook that we looked at with the inner spersed comments and pulls out just the Julia code so right click on the code open an integrated terminal pretty cool that cursor enables even one computer to do stuff and and a lot of externalities on the compute so so you know local there's a lot of cool ways to do this um but using multiple different Cloud resources like this hope that it's all useful and relevant in the broader service of of living and and computation start out with just running it as it first converts Julia drone Dynamics it may need a slight reordering or it could just be like my environment or we'll see um one feature that comes up a lot in the um RX and fur group and and then and learning about it is just like this very Julia ask you usage of statistical variables very close to as they are analytically isomorphically uh I mean ideally and and um learning the syntax of what is allowed sort of in in in plausibility where is it speculative realism distributional uh specification where and how does the does the current ability to make different kinds of nodes and Implement different kinds of edges in RX and fur which could be more uh general or robust in the future I'd expect so just from an open source software development side like what kinds of app models will what kinds of statistical Behavior arise from what kinds of Logics and functions like go to Target are being um put within the app model what's in the logic of updating and calling different nodes there's just a lot of interesting areas and uh spaces for for taking these functional examples and if if and being clear about where where are we having problems in implementing these examples like I don't expected to run the first one because it was a notebook but then it hopefully is just one or a few prompts away and then we can start taking these things apart looking at making more logging um testing different variants please update this to run it just hangs it was converted from notebook break it up out into any separable modules properly invoked if you think it best okay it looks possibly let's just see what happens if we say move ahead okay let's this one seems to be hanging again so here one one affordance that they sort of point two is new chat so this is back to open manace here is what we get now use a chain of open manace calls and Tool use on incrementally writing reports and doing dispatch of document writing then stitching together as editor at the end okay this is still happening so that took 1,800 seconds okay it's loading some information this is like this is getting into the um yeah so stuff is coming out but this is a one of the big open questions that hopefully we won't get too off the rails with in this with the tools but even with the um and perhaps even especially with the H Hazard and just open source grabbing these tools but even if it were a much more well orchestrated documented approach even that would be be an especially important like what is really being empowered in the environment that I'm running this in um where are different kinds of real consequences possible it's just um it's pretty interesting and and how does that relate to the actual technical basis of what the model is and does and how it's interpreted okay some drone core we can use perplexity to make a drone core image see if we can get the studio jii okay yeah so in the time when it wrote the 17 Page PDF so it didn't output uh the other discourses and but a lot of more stuff was happening but it it just saved in in the time that I previously ran this just 17 Page PDF on this thermal PDP model and concept but it really did have files like this file partially haphazardly outside of its uh Lane but not um not few files were were made hugging phase data sets really may be coming here there really were Python scripts that did have some outputs and some source code was written so we'll see where where that goes and I think that's another sort of General topic is just file structure handling to what extent should more relational databases and uh General High Performance databases and maybe they're already being used in in in different ways to make some of the handling of the reference of these different system components more facilitated doing it locally like all of these examples has a lot of interpretability it's like very Hands-On but for high reliability there's some other methods but those does have their own sort of tradeoffs so okay so we got open Manis being improved by cursor to make more sequential incremental State sharing amongst runs um let's just run the main this is um uh let's let's run this is the initialized script activate the virtual environment yes run open manness yes okay oh okay okay drone example

March is on pop out terminal so it is happening could be hanging in a really useless place okay code execution in the agent Laboratory test train split loading some data from hugging face lab number two is the index of that lab so it's like where I I don't know where this is happening in the in the cache so where do different files get stored where does that get stored in databases Etc okay looks like some kind of error but what what was very interesting to see is this so hm this is a just wasn't sure what level of code this even was in terms of Errors happening in certain tasks blurring with written scripts because the scripts for the tasks are in this repo which also has Eda access to these tools and to cursor so it's just it's probably not the um most crystallized High reliability setting but it has this sort of open-ended Evolution element with the agent laboratory especially um okay open Manis common cursor pathology let's let's let's do something totally different use the open let's just start a new chat fix this so that main. pi Works fully functionally as it should okay RX infer drone model is running post do just being post dos 2D okay while that's running add more logging to the outputting so we better with emoi structuring understand and and can report on the timing and resource uh just timing of different okay so RX Ander at least is doing some logging here of running the Drone Dynamics simulation break out the core program spec and logic flow from the statistical description of the G at model stat okay errors from Julia so now with the errors from the logging from it fix this and add more there open Manis okay now main is working all right so here's what the open man is functionality what just running the main enter your prompt I am in a live stream on multi-agent Dynamics research and report on with five paragraph essays code stories about what kinds of State ofth art methods it would make sense comprehensively understand in April 2025 oh already let's see what time it is okay let the open manace repair still writing PhD dialogue now it's into the storying of the um all right back to the Drone working on drone Dynamics so hopefully that even even though I'll do about 10 or 15 more minutes um these are all right in the zone of just hit and miss right now and so I'm just sort of that's why I'm showing from a more reset State um but I think people can report back and and obviously we will see um what is really possible with making these systems a lot more High reliability in in multiple key ways so okay so yeah this is coming from open manace this is with the browser so let's pop this out yeah incredible okay it popped up this window with these analysis boxes brought here is going to be the terminal this is man is selecting these tools is it the uh you know is it the best possible way to look at this action selection or you know how to interact best with sites with the some many degrees of freedom but that's a pretty cool out outcome so let's let that browse and and and consider the watershed moment while agent laboratory is writing the thermal PDP and now we have the improved logging RX infer drone form let's do another drone script same total functionality with fewer time steps well let's just look let let's reduce the number of time in our simulation okay oh add to chat and fix this error so here's the Step 11 so that's what that was the chain of of tool use that was leading to that website the news website okay all right it has popped up family center. met. looks like possibly SL Brazil in Portuguese unexpected but okay okay grab attention again with that paper continuing on in the open AI variably successful um modification using LLS M of RX infer code so that's one one reason it's fun to do the script first off it allows a lot more articulation and separation between different parts of the code better logging for for when things are not working it can break things too but that's the helper script so as long as they keep the main RX infer examples notebook functional or or similar reference models of code then it's a lot easier to regenerate functional scripts and start modifying from there yeah I'm going to do about 10 more so if anyone has any comments or questions they want to have mentioned in the last 10 minutes write them okay let's look at how tool use really output okay running just 2D quick so let's see if this will successfully save an output it was working a little more earlier on with the the numbers and the outputs so let's see if we can at least recover that if not save into but once we have the scripts and the ability to to modularly break it out in pretty simple steps then there can be different logging and saving it can be okay well now just deposit that Vector into this Matrix form and then just keep on working from there so there's a lot of ways to use even just the um State space Clarity that making these probabilistic models can provide that's kind of the systems modeling part that's that's always really interesting so 15 Steps deep possibly just what did we even ask for what time is it Overkill right but General so open man is probably not not even pretending to be otherwise than such seems like a little bit like less of a functional version it's it's but it's it's um a great direction to um at least point in that direction agent laboratory has the associated paper let's hope that it will finish in the last again things are happening with open Mana so it has that sort of tool architecture paper still writing over there 16 out of 20 open Manas drone error yeah and also it's helpful to have the notebook form available in the code base so then it can be referenced say okay this was working especially early on yeah I feel that this will not finish and render PDF but the the PDF that it did render previously was keeps like going to this website let's see what one it opens next the the the PDF was styed probably one or two times but then generally was rendering fine so they used a a bunch of different methods it's really cool work um yeah multi-agent acto blockprints where we've for for a few years sort of with es and flows talked a lot about these

these topics about uh approaches for cognitive modeling ways to take procedural approaches ontologically driven approaches to to modeling connecting that with data Integrity stuff like from block science with different and other requirements engineering P3 these kinds of modeling layers oh it detected that the browser window is closed maybe that shouldn't be done next time let's see if it can fix the Drone example in the next 7 minutes but if anyone has any last comments go for it okay close that oh yeah the Drone core images and perplexity not sure why they kind of broke it out it was nice when it was on that side tab um gave a talk earlier today and this was a one of the themes was the two sides of this coin natural foraging systems and Engineering forging systems and algorithms and this sort of bidirectional relation this perspective swap the systems as they are and the empirical data the the inspiration they provide and then also these computer scientific more engineered or more mathematical uh models multi-agent settings and in the ways that those are kind of coming together giving all these different optional for modeling okay see stunning s only accessible by rail okay drone at least is started up this one always looks like it's halting but I think at these these stages maybe when it runs the um paper solver even just one time this could be a lot faster probably with a parallelized labs and with uh not just this older laptop that I'm running this from um this this seems like there's a lot of backend um possibilities and from the requirements installation like there are some packages so like huh wonder why they're doing that and so it seems like there's some pretty um Advanced methods that are being used sequentially and that was what I sort of broke it trying to get at and and but what I know will will be part of it in in the bigger picture is that kind of expressivity graphical user interface from no to low to very medium and high code High touch ways of working with informations these are all just sort of again halfhazard archetypes and um they will already in their own description of frictions and just looking at functions here as randomly assembled on the given afternoon um and pointed to by these repos and peers and sort of adjacent from from this sort of concept there's just a lot of space to think about how how to get this to all work in whether this is the best agent orchestration framework or as several times I've just built it from scratch with just a few prompts or whether using some other agent orchestration Frameworks in any given moment that's that's a super important question and I I hope that we continue to develop down those lines and stay multiple there um there's just sort of overlapping and complimentary functions that difference modeling approaches can yield and uh having the the modeling capabilities connected with like requirements engineering pattern languages those kinds of features so that there can be formal and natural language um effective descriptions of systems connecting that to the compositionality of multi-agent LM browser type code writing and uh reviewing improving of the code which at least the first time that I got it to work it it totally did do sequential code Improvement and running I don't know with what resourcing it was running the code or how it was getting to the outputs of the code um in terms of its workspace and all of that but but but those are just the questions um 20 steps or what what's the right number of what compositions content empty so even though while while the outputs are in many ways less impressive than just one perplexity um search might get you it's about these functionalities and different composition alties being more and more accessible more multimodal um part of larger orchestration Frameworks more open to calling and receiving API calls and different information measures on the input and the output and the cognitive nature of systems um that sort of low road High Road um whether it's as systems uh must be for a given Observer all these different ways that people talk about the different um relational stance and pragmatism how that's all connecting with just the ability to Vibe code and uh make documentation tests logging functions a lot of different things that okay this this finish did it even save a file maybe but it did do some browser opening so okay goodbye open Menace so now we have just in this last minute see what happens we have the uh lab which does the and and and prior did get some artifacts being built of just multi-agent coordination and the RX andur giving some issues for the Drone example but was much easier to move over to the script form for the neural network and the recurrent uh switching linear dynamical systems model so thank you all hope that was interesting Andor useful not as much baseball and so on as there could have been but say lovey so thank you till next time bye